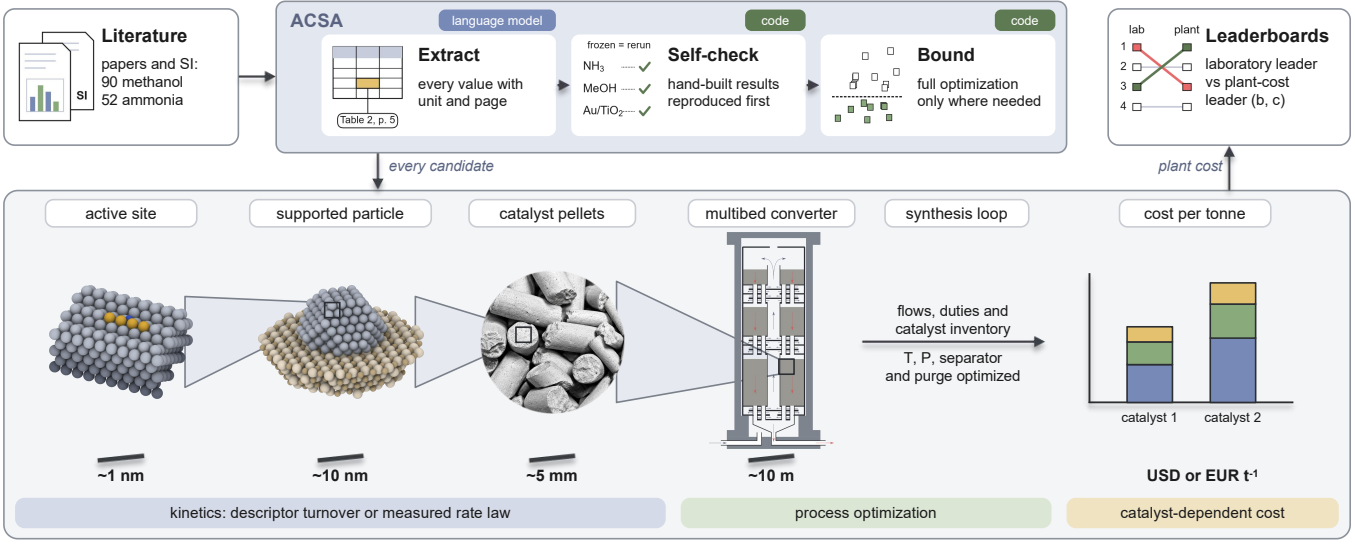

Literature
papers and SI:
90 methanol
52 ammonia
SI
ACSA
language model
code
code
Extract
every value with
unit and page
Table 2, p. 5
frozen = rerun
NH3
MeOH
Au/TiO2
Self-check
hand-built results
reproduced first
Bound
full optimization
only where needed
lab
plant
1
2
3
4
Leaderboards
laboratory leader
vs plant-cost
leader (b, c)
every candidate
plant cost
active site
supported particle
catalyst pellets
multibed converter
synthesis loop
cost per tonne
flows, duties and
catalyst inventory
T, P, separator
and purge optimized
catalyst 1
catalyst 2
~1 nm
~10 nm
~5 mm
~10 m
USD or EUR t-1
kinetics: descriptor turnover or measured rate law
process optimization
catalyst-dependent cost